

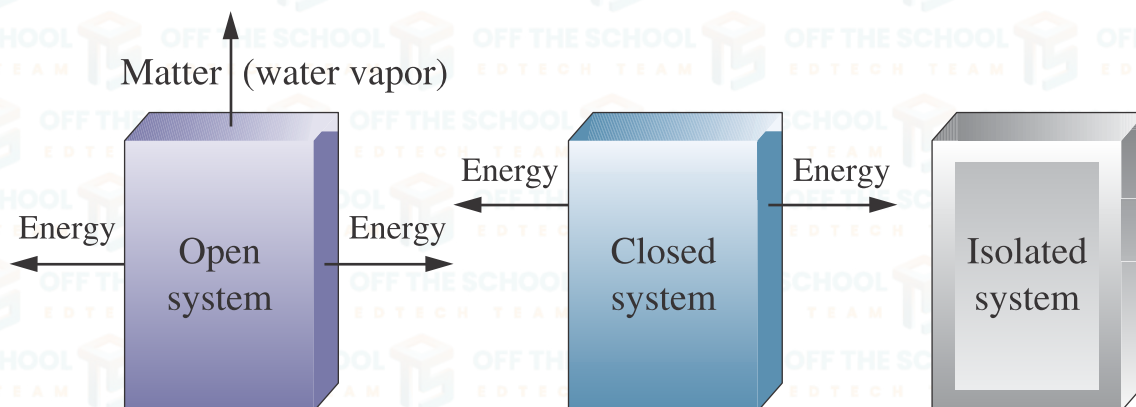
UNIT 11

THERMOCHEMISTRY

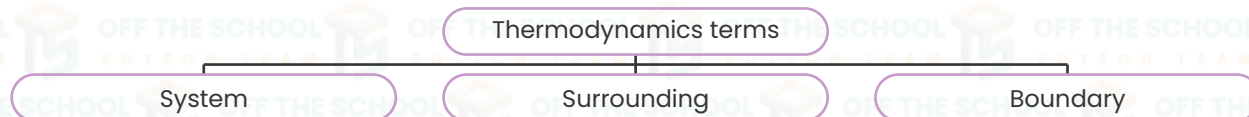


THERMOCHEMISTRY:

The study of energy transference as heat during a chemical process. Chemical reactions are generally carried out through the transfer of energy to or from the surrounding, the study of which is known as thermochemistry.

**THERMODYNAMICS:**

The branch of science deals with the study of the inter-transformation of heat, work, & other forms of energy. Thermodynamics deal with the macroscopic properties of the system (that is it concerned with the initial & final states of the physical or chemical process) & doesn't tell about the molecular constitution of the system.

THERMODYNAMICS TERMS:**SYSTEM:**

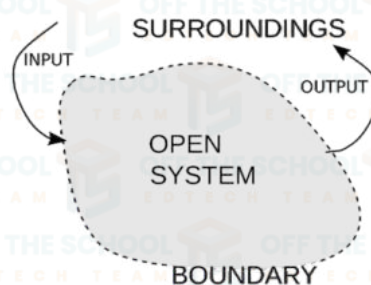
The distinct part of the universe which is under experimental observation & separated from its surrounding by an imaginary or real surface & change its shape or size.

SURROUNDING:

Anything outside the arbitrarily selected boundary of the system that affects the system directly or indirectly.

BOUNDARY:

It is a physical (real) or conceptual (imaginary) closed surface that separates the system from its surroundings.

**EXPLANATION THROUGH EXAMPLE:**

- The part of the thermodynamic universe under observation is the system while the remaining part which affects the system is the surrounding & the region of distinction between the two is the boundary.
- Consider a reaction between $\text{Ba}(\text{OH})_2$ & $2\text{NH}_4\text{NO}_3$ carried out in a beaker.



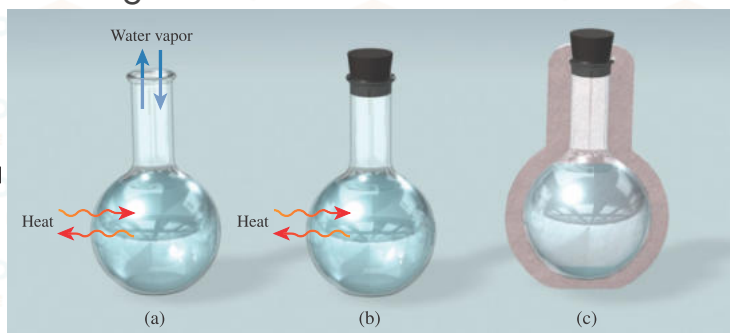
Illustration of a thermodynamic system The system consists of the portion of the universe that we choose to study; in this case, it is a solution of $\text{Ba}(\text{OH})_2$ and NH_4NO_3 and the flask. Everything else constitutes the surroundings.

- The reaction constitutes the system & characterized by mass, volume & temperature, etc.
- The remaining part includes the walls of a beaker, and the empty part of the container & workspace is designated as the surrounding."
- The walls of the beaker are physical boundaries.

THERMODYNAMICS SYSTEMS:

Based on mass & energy transfer between the system & surroundings, thermodynamic systems can be categorized as:

- (a) An open system allows exchange of both energy and matter with the surroundings. (b) A closed system allows exchange of energy but not matter. (c) An isolated system does not allow exchange of energy or matter. (This flask is enclosed by an insulating vacuum jacket.)



OPEN SYSTEM:

All thermodynamic systems through which both material & energy can be transferred are called an open system.

- The boundary of such a system is open & non-insulated.
- **Example:** boiling tea in an open pot.

CLOSED SYSTEM:

All thermodynamic systems through which only energy can be transferred but the material cannot be transferred are called closed system.

- The boundary of such a system is sealed but non-insulated.
- **Example:** boiling tea in the pressure cooker.

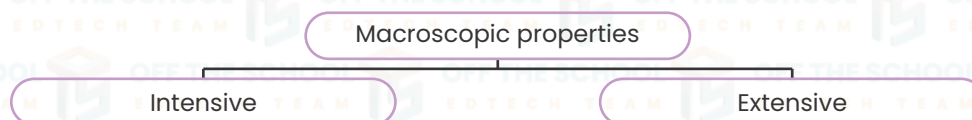
ISOLATED SYSTEM:

All thermodynamic systems through which neither energy can be transferred nor material can be transferred are called isolated system.

- The boundary of such a system is sealed & insulated.
- **Example:** boiling tea kept in a thermos flask.

MACROSCOPIC PROPERTIES:

Thermodynamic properties of the system in bulk amount, that depends upon initial & final state of process & doesn't tell about the molecular constitution of the system.



INTENSIVE PROPERTIES:

The properties of the system which is independent of mass are called intensive properties.

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Example: Density, Vapor pressure, temperature, viscosity, surface tension, refractive index, melting point, boiling points etc.

EXTENSIVE PROPERTIES:

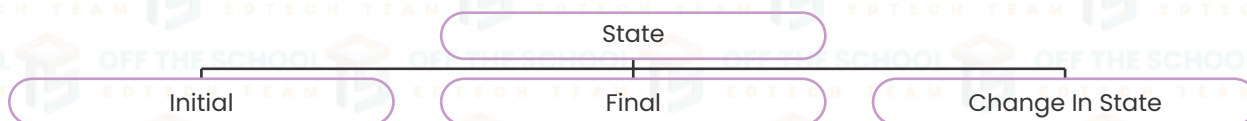
The properties of the system which is dependent on mass are called extensive properties



Example: Mass, volume, moles, enthalpy, entropy; internal energy, free energy, etc.

STATE:

The thermodynamic system is said to be in a certain state if some macroscopic properties (temperature, pressure, volume, moles) are known.



INITIAL STATE:

The description of properties of the system before the change occurs.

FINAL STATE:

The description of properties of the system after the change occurs.

STATE CHANGE:

The change in macroscopic properties of the system from the initial to the final state.

Example: Initial temperature of water is T_1 , on boiling its final temperature is T_2 , change in state (temperature of water is $\Delta T = T_2 - T_1$).

STATE FUNCTION:

A thermodynamic parameter that is dependent on the initial & final state of the system & does not bother with how it happens.

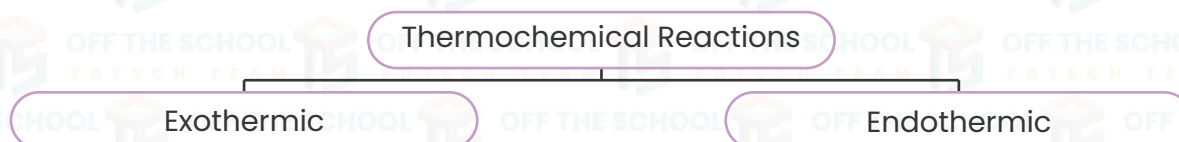
Example: Examples of state functions are internal energy, enthalpy, entropy, volume, and temperature.

The change in elevation that occurs when a person goes from the ground floor to the fourth floor in a building does not depend on the path taken.



THERMOCHEMICAL REACTIONS:

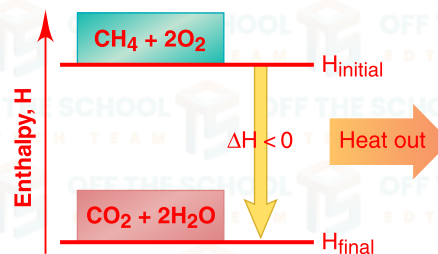
The chemical reaction which are proceeded by the evolution or absorption of heat energy is called thermochemical reactions.



EXOTHERMIC REACTIONS:

The chemical reactions which are proceeded by the evolution of heat energy are called exothermic reactions.

- In exothermic reactions, heat energy is released which makes the surroundings warmer.
- The heat contents of the products are less than the heat contents of reactants.
- Enthalpy change is negative ($\Delta H < 0$).
- The reactions are energetically favorable.
- Usually, these reactions are spontaneous.

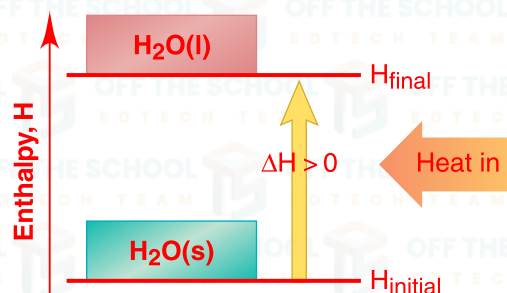
**A Exothermic process****EXAMPLES:**

- Combustion of hydrocarbon:

**ENDOTHERMIC REACTIONS:**

The chemical reactions which proceed by the absorption of heat energy are called endothermic reactions.

- In endothermic reactions, heat energy is absorbed which makes the surroundings colder.
- The heat contents of the products are greater than the heat contents of reactants.
- Enthalpy change is positive ($\Delta H > 0$).
- The reactions are not energetically favorable.
- Usually, these reactions are non-spontaneous.

**B Endothermic process****EXAMPLES:**

Conversion of water into steam:

**FACTORS AFFECTING THERMO CHEMICAL REACTION:**

The following are the factors that affect thermo chemical reactions:

1) Amount of reactants:

The larger the number of chemical reactants larger will be the heat evolved or absorbed during the chemical reaction.

2) Ions occur at constant volume:

For the reaction occurs at constant volume,

"The evolution or absorption of heat energy during the chemical reactions occurs at constant volume is equal to the change in the internal energy of the system"

$$Q_v = \Delta E$$

3) Action occurs at constant pressure:

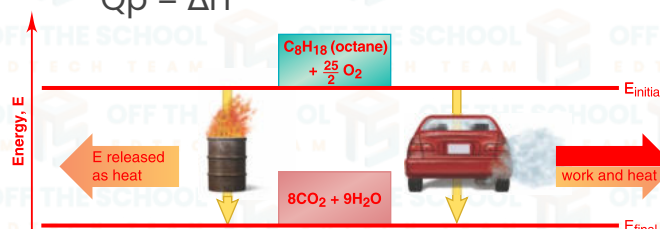
For the reaction occurs at constant pressure,

"The evolution or absorption of heat energy during the chemical reaction occurs at constant pressure is equal to the change in the enthalpy of the system"

$$Q_p = \Delta H$$

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Two different paths for the energy change of a system. Even though q and w for the two paths are different, ΔE is the same.



4) Temperature:

Temperature also affects thermo chemical reactions depending upon the fact the temperature change is favorable or not.

1st LAW OF THERMODYNAMICS:

Introduced by: J.R. Meyer (1841), **Formulated by:** Helmholtz (1847)

STATEMENT:

In any physical or chemical process, the net energy of the system & surroundings must remain constant

Or

A system can't create or destroy energy, but it can exchange energy with its surrounding in the form of heat & work

EXPLANATION:

- The first law of thermodynamics is based on the law of conservation of energy.
- According to this law, heat & work, are mutually inter-convertible in such a way that the total energy of the system & surrounding always remains constant.

MATHEMATICAL FORM:

Let, E_1 is the initial internal energy, if q heat is supplied to the system & W is the work done on the system, the internal energy increased to E_2 .

According to this law, internal energy change ΔE can be formulated as:

$$\Delta E = q + w$$

where, q is heat transfer across the system W is work done ΔE is the change in internal energy of the system

HEAT TRANSFER WITHOUT WORK DONE:

$$\Delta E = q + w$$

$$\text{But, } W = 0$$

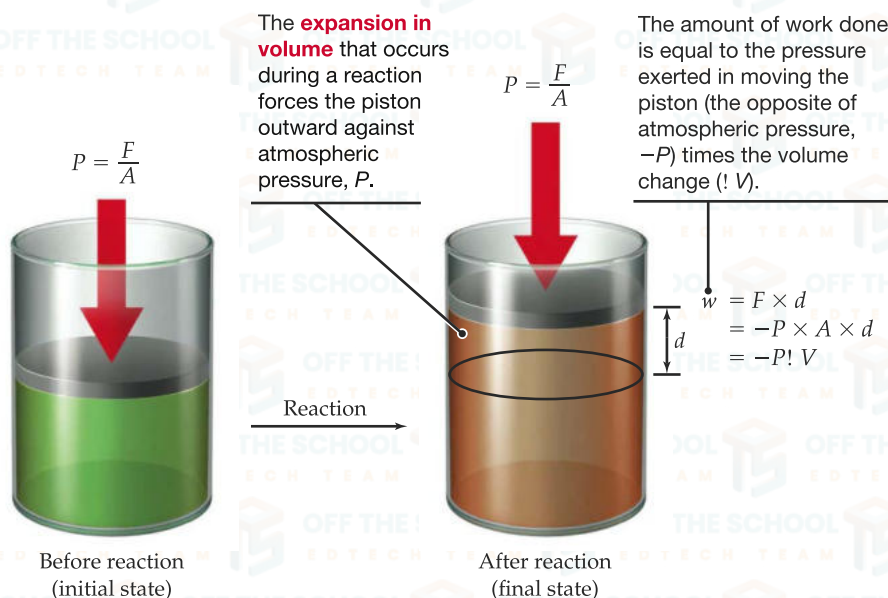
$$\Delta E = q$$

WORK DONE WITHOUT HEAT TRANSFER:

$$\Delta E = q + w$$

$$\text{But, } q = 0$$

$$\Delta E = w$$



INTERNAL ENERGY:

The total energy contents of thermodynamic system under a given conditions is called internal energy.

EXPLANATION:

- In any instance, a system composed of particles (atoms, molecules, or ions) has a specific amount of energy.
- Kinetic energy is due to various motions (translation, vibrational, rotational) of particles.
- Potential energy is due to attractions & repulsion between particles or as energies stored in the chemical bond.
- All these energies are collectively termed as internal energy of the system.

ENTROPY:

The measure of the disorder of the system.

Example: The entropy of gases is more than liquids due to the free arrangements of molecules in the gas.

ENTHALPY:

The total heat contents of the system are equal to the sum of internal energy & algebraic product of pressure & volume. $H = E + PV$

Enthalpy is the state function composed of all state functions E = internal energy, P = pressure & V = volume.

Units: Enthalpy is measured in Joules (J) or calories (Cal) which are interrelated as:

$$1 \text{ Cal} = 4.184 \text{ J}$$

In chemical reactions amount of heat absorbs or released is expressed as Enthalpy with unit kJ/mol.

**THE ONE WHO WAITS GETS ONLY
WHAT THE ONE WHO TRIES
COULDN'T GET.**

—(DR. ABDUL QADEER KHAN)—

